

EXTERNAL HARD DRIVES

When it comes to sharing your data or simply synchronising data between your office and home, external hard drives play a vital role. The best thing is that you do not need to open your cabinet to transfer data—just plug it into the USB port and play!

We selected 13 drives for this test. Of these, two were from Iomega, five from WD, three from Maxtor, one from Philips, and two were from Seagate.

Backup External Hard Drives

These drives are meant for large and frequent backups. Capacities start from 250 GB, and the prices are considerably higher, too. Most of them come with backup software of some kind and special features such as backup buttons. FireWire ports are common too. These drives are bulkier than those in the portable drive category (read on) as they seldom need to be carried around. System administrators, rather than home users, are the consumers for this category of external hard drives.

There was a 250 GB Iomega, a 500 GB and a 1 TB drive from Maxtor, three 500 GB and one 1 TB drive from WD, a 250 GB drive from Philips, and a 750 GB drive from Seagate in this category.

Features

Capacities and RPM

Capacities in this category ranged from 250 GB to 1 TB. The 1 TB drives were the Western Digital My Book Pro Edition II and the Maxtor OneTouch III Turbo Edition, containing two hard drives each. Being backup drives, a higher



Western Digital My Book Essential
Great value backup!

capacity is always preferred, considering the fact that even entry-level PCs come with over 100 GB drives today. All the drives had an rpm of 7200, and most featured a healthy 16 MB buffer, which is adequate for just about any purpose.

Interfaces

All the drives except the Iomega, Philips, and WD My Book Essential featured the USB 2.0 as well as the FireWire interfaces. The above-mentioned drives featured only USB 2.0. The other drives came with two FireWire ports, which lets you connect more drives in a daisy-chain, and does not require the computer to have more FireWire ports to add compatible devices. This can be used to either connect other FireWire devices such as DV cams, or to augment drive space by connecting more such drives. Some drives such as the Maxtor drives and the WD My Book Pro Edition and Pro Edition II even had FireWire 800, which supports a data throughput of 800 Mbps—twice of that offered by FireWire 400.

Bundled Software

EMC Retrospect was the software of choice of most brands. It came with the Maxtor drives, the Iomega, as well as the Western Digital My Book Pro Edition and Pro Edition II. This is a premium software that allows you to back up as well as synchronise your data with your laptop or PC. The WD My Book Premium Edition came with WD Backup, a proprietary backup software, while the WD My Book Essential came with no backup software at all.

Advantage SATA

SATA is the de facto standard of hard disk storage today; the older IDE or PATA is *passé*. So what's different? Does it deliver what it promises? Is it really worth it or is it hype?

To start with, SATA drives support a faster data throughput than IDE drives. While IDE drives cap at 133 MBps (1.04 Gbps), SATA drives begin at 1.5 Gbps, while SATA II (actually, a misnomer—it should be called SATA 3.0 Gbps) can go up to, well, 3 Gbps. When benchmarked against IDE drives, it does show an appreciable improvement in the throughputs; these numbers are therefore not just on paper—you can see the performance improvement in real-life scenarios such as file copying and application loading.

SATA drives feature a four-conductor data cable that does not just make it slimmer, remain tangle-free, and enable better ventilation in the cabinet, but also a lot easier to attach.

Hot-plug is yet another advantage offered by SATA drives. This feature lets you plug or remove a SATA drive without switching off the PC. The SATA controller on the motherboard must support this feature, though.

Another important feature of SATA drives is Native Command Queuing (NCQ), which it borrows from SCSI drives. NCQ improves efficiency and performance. During normal reads or writes, commands arrive at the disk and are executed on a first-come-first-served basis. This creates a mechanical overhead when the read/write head has to be constantly repositioned. SATA drives make use of an algorithm to

determine the most efficient way to execute this queue of commands to create the least possible mechanical overhead. This results in a general performance increase. This feature was initially only found on SATA II drives, but has since been incorporated in many SATA I drives too.

The Port Multiplier spec on SATA II allows you to connect up to 15 such drives to a single SATA controller via a port multiplier. Though this still compares unfavourably when you consider the number of drives that can be connected to Fibre Channel or Serial Attached SCSI (SAS), it will make it easier to build disk enclosures for SATA drives.

Port Selectors allows two hosts to be connected to a drive, thus creating a redundant connection to the drive. Thus, if one of the hosts fails, the other can take over and the PC will not go down. A home user will not be much interested in this feature, but the enterprise segment would find this very useful.

As you can see in this shootout, which featured a few SATA I (1.5 Gbps) drives as well, there is not too much of a performance gain with SATA II. Do not, therefore, expect miracles from these drives; they are just a tad bit faster in real-world tests.

The prices of both versions are the same, and further, most drives are backward-compatible with SATA I controllers. In case of any compatibility issues, you can even manually set the SATA II drive to function as a SATA I drive. It therefore makes sense to go with these drives rather than SATA I.